Imbalance Mitigation for TCGA-UT Across Patch and WSI-Bag Regimes

Abstract

Class imbalance in computational pathology is not a single evaluation problem: labeled image patches support ordinary image classification, whereas whole-slide image (WSI) classification is commonly formulated as weakly supervised multiple-instance learning over patch-feature bags. This study evaluates imbalance mitigation on TCGA-UT with two native-regime benchmarks rather than a single cross-regime leaderboard. The patch benchmark compares cross-entropy, class weighting, focal loss, balanced sampling, Scholz-style combined losses (CE with soft F1 or soft MCC plus balanced sampling), and ProGAN-based synthetic augmentation. The WSI-bag benchmark compares matched attention-MIL baselines with RankMix and SC-MIL. All methods use slide-disjoint splits, repeated three-seed evaluation, and shared backbones within each benchmark. On patches, plain cross-entropy gives the highest macro F1 (0.314 ± 0.004), while balanced sampling gives the highest balanced accuracy (0.333 ± 0.013) at lower macro F1. On WSI bags, RankMix is the strongest method, improving balanced accuracy over plain MIL CE from 0.836 ± 0.020 to 0.863 ± 0.008 and macro F1 from 0.840 ± 0.020 to 0.848 ± 0.017. The results show that advanced imbalance mechanisms are not uniformly beneficial, but can help when their assumptions match the native input regime.

1 Introduction

Class imbalance changes what a pathology classifier is rewarded for learning. When common tumor types dominate training data, high aggregate accuracy can coexist with unreliable rare-class recall, especially if evaluation emphasizes prevalence-sensitive summaries. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware summaries as primary outcomes in imbalanced settings (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

A common source of ambiguity in pathology imbalance studies is the input regime. Some methods are naturally image-level methods: they assume labeled images, image embeddings, or directly generated image samples. Other methods are naturally WSI-level methods: they assume weak labels over bags of instances and use attention, instance ranking, or bag-level representation learning. Comparing such methods in one pooled leaderboard can confound the imbalance mechanism with changes in supervision, input size, backbone, and aggregation.

TCGA-UT is useful for separating these questions because it supports two legitimate tasks with different native inputs. The dataset provides labeled histopathology patches cropped from pathologist-selected tumor regions and slide identities that can be grouped into WSI-level bags. Patch labels are class labels, not dense pixel masks or cell-level annotations. This study therefore uses the available supervision directly: image-level methods are evaluated on controlled patch classification, while MIL methods are evaluated on slide-level feature bags.

The contribution is a controlled benchmark design rather than a reproduction claim for every cited method. The study uses one dataset, two native tasks, and one controlled comparison within each task. Method fidelity is defined by implementing the defining imbalance mechanism in the regime for which that mechanism was designed, while holding the backbone family, split, and evaluation code fixed within each benchmark.

2 Dataset

TCGA-UT is derived from diagnostic *whole-slide images* (WSIs) in The Cancer Genome Atlas (TCGA), a pan-cancer resource that links molecular and clinical cancer data with large image collections (Tomczak et al., 2015). The TCGA-UT release turns this slide archive into a computational-pathology image dataset by extracting hematoxylin and eosin (H&E) tumor *patches* from pathologist-selected *uniform tumor regions*. It contains 1,608,060 RGB patches of size 256×256 pixels from 8736 diagnostic slides and 32 *cancer-type labels*. The patch labels are slide- or cancer-type labels, not dense tissue masks or cell-level annotations (Komura et al., 2022).

This structure makes TCGA-UT well suited for an imbalance study because the imbalance is native to the dataset rather than imposed only after sampling. The largest class is glioblastoma multiforme (GBM; 792 slides), while the smallest class is diffuse large B-cell lymphoma (DLBC; 28 slides). The resulting slide-level imbalance ratio is 28.3. As a consequence, the dataset exposes the practical tension that motivates this study: methods must learn from many visually related cancer categories while preserving performance on classes with far fewer slide-level examples.

Table 1: Dataset summary used by the shared patch and WSI-bag benchmarks.

Property	Value
Cancer-type labels	32
Diagnostic slides	8736
TCGA-UT patches	1,608,060 patches of 256×256 pixels
Largest class	Glioblastoma multiforme (GBM), 792 slides
Smallest class	Diffuse large B-cell lymphoma (DLBC), 28 slides
Slide-level imbalance ratio	28.3
Shared split size per seed	6116 train, 1310 validation, 1310 test slides
Controlled patch input	Resolution level 0, up to 30 deterministic patches per slide
WSI-bag input	Full frozen Virchow2 feature bags from the same slide identities

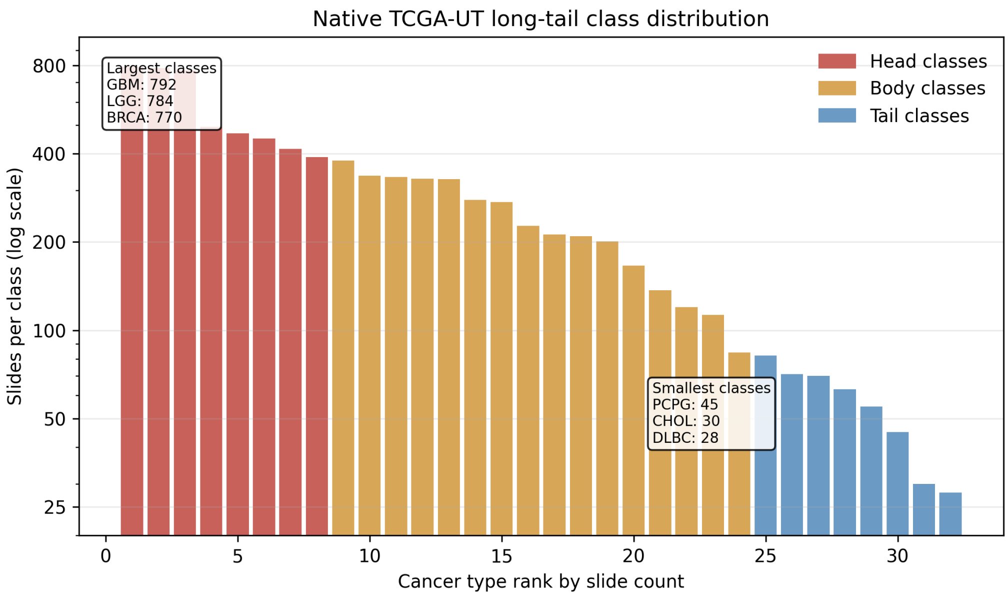


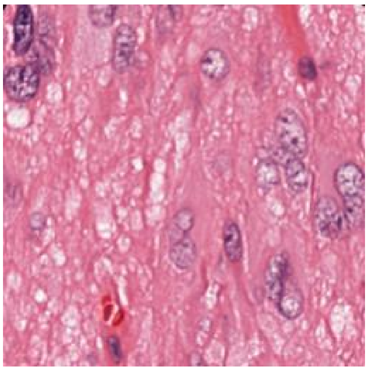
Figure 1: Native TCGA-UT slide support by cancer-type rank. The long-tailed distribution motivates metrics that weight classes more evenly than aggregate accuracy.

2.1 Data Units and Terminology

Several data units are involved in the benchmark, and the distinction matters for method comparison. A *slide* or *WSI* is one diagnostic histopathology image associated with one cancer-type label. A *tumor region* is a pathologist-selected area of a slide from which TCGA-UT crops groups of patches. A *patch* is one 256×256 RGB crop used directly by the patch classifier. A *feature chunk* is a saved frozen-feature representation associated with a slide or slide subset. A *WSI bag* is the collection of frozen patch-feature instances used by the attention-MIL model for one slide-level prediction. *Head*, *body*, and *tail* classes refer to relative class-support tiers in the long-tailed slide distribution. A *slide-disjoint split* means that all patches and features from one slide stay in exactly one of train, validation, or test.

Table 2: Terminology used for the two native input regimes.

Term	Meaning in this study
Slide / WSI	Diagnostic whole-slide image carrying one cancer-type label.
Patch	256×256 H&E crop from a selected tumor region; patch labels inherit the slide cancer type.
Region	Tumor area selected before patch extraction.
Cancer type	The 32-way class label used for patch and slide-level classification.
Feature chunk	Frozen Virchow2 feature representation used to assemble slide-level feature bags.
WSI bag	Set of patch-feature instances for one slide, consumed by the MIL model.
Head/body/tail class	Class-support tier derived from the slide-count distribution.
Slide-disjoint split	Split rule preventing any slide's patches or features from crossing train, validation, and test.



Field	Example value
Cancer type	Glioblastoma multiforme
Resolution level	0
Patch size	256×256 RGB pixels
Source unit	One crop from a selected tumor region
Patch role	Image-level sample if selected by the controlled manifest
WSI-bag role	Candidate instance in the slide's feature bag

Figure 2: Example TCGA-UT data point copied from the shared raw dataset. The image is a single patch, while the table shows how that patch relates to the manifest fields used by the benchmark.

2.2 Shared Splits and Benchmark Inputs

For each seed, slides are split into 6116 training, 1310 validation, and 1310 test slides. The same slide-level assignments are reused by both benchmarks so that no patches from the same slide can appear in different splits. This is the main leakage-control mechanism: the patch benchmark and the WSI-bag benchmark see different input units, but they inherit the same train/validation/test slide partition.

The patch benchmark derives a deterministic controlled patch manifest from the shared slide split. It uses resolution level 0 and samples up to 30 patches per slide with a fixed rule before method-specific training begins. The cap is a compromise between data coverage and slide-level control: it uses the empirical median number of raw JPG patches per slide at resolution level 0 (median 30, mean 31.1, range 10–80) while preventing slides with many extracted tumor-region patches from dominating the patch-level loss. At this resolution, 8686 of 8736 slides have at least 30 raw patches, so the cap retains the typical per-slide patch evidence while limiting only the larger slide folders. This makes the patch benchmark a controlled comparison of imbalance mechanisms rather than a comparison of patch-selection policies. The WSI-bag benchmark uses full frozen Virchow2 feature bags built from the same slide identities, rather than applying an additional instance cap (Zimmermann et al., 2024).

3 Methods

The methods are organized by the level at which they intervene in the imbalanced learning problem. Cross-entropy (CE) is treated as the no-mitigation reference condition: it uses the observed training distribution as given and does not rebalance samples, labels, costs, representations, or generated data. The remaining methods are tested imbalance interventions grouped according to the taxonomy used in the class-imbalance literature: data-level sampling and augmentation, algorithm-level losses, hybrid sampling–loss methods, synthetic generation, and representation or architecture-level methods (Saini and Susan, 2023; Zhang et al., 2023).

The comparison is regime-specific. Patch methods consume individual RGB patches and use one shared patch-classification backbone. WSI-bag methods consume frozen Virchow2 feature bags and use one shared attention-MIL backbone. Thus, method differences within a benchmark correspond to imbalance mechanisms rather than changes in input unit, split, or evaluation code.

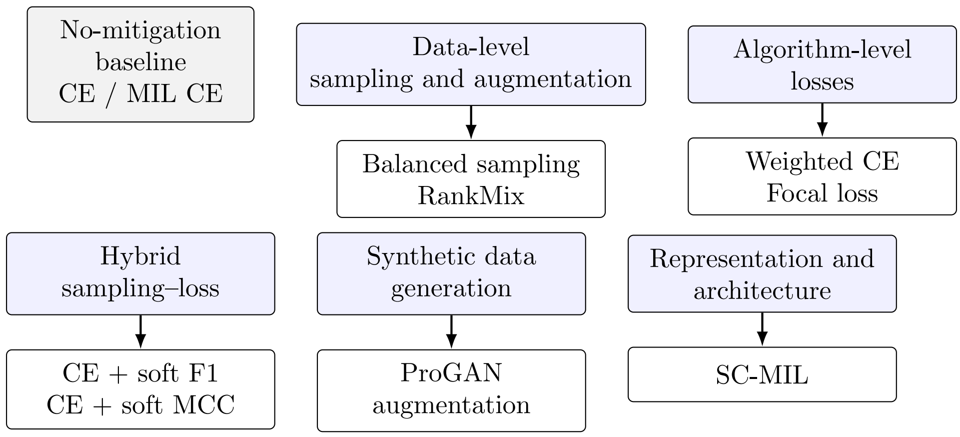


Figure 3: Method organization used in the benchmark. Cross-entropy is the no-mitigation reference, while the tested alternatives intervene through sampling, loss design, synthetic data, WSI feature augmentation, or representation learning.

3.1 Notation

Let C be the number of cancer-type classes and let n_c denote the number of training examples from class c. For a labeled patch or slide-level bag with label $y \in \{1, \dots, C\}$, the model outputs logits $\mathbf{z} \in \mathbb{R}^C$ and class probabilities

$$p_c = \frac{\exp(z_c)}{\sum_{j=1}^{C} \exp(z_j)}. \tag{1}$$

For WSI-bag methods, a slide is represented as a bag $B = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ of frozen patch-feature instances. The attention-MIL backbone maps each bag to a bag embedding $\mathbf{b}(B)$ and class logits. Unless a method explicitly changes the sampler, objective, or representation loss, the empirical class frequencies in the training set define the optimization signal.

3.2 Cross-Entropy Baseline

Unweighted cross-entropy is the baseline because it is the standard supervised objective without an explicit class-imbalance correction. For an example with label y, the loss is

$$\mathcal{L}_{\mathrm{CE}}(\mathbf{z}, y) = -\log p_y. \tag{2}$$

In the patch benchmark, this objective trains the shared patch classifier on the controlled patch manifest. In the WSI-bag benchmark, the same objective trains the shared attention-MIL model on slide-level feature bags. In both cases, CE uses the observed class distribution directly: majority classes appear more often in minibatches and therefore contribute more loss terms over an epoch. This makes CE an appropriate reference for asking whether an imbalance-specific intervention actually improves the rare-class tradeoff.

3.3 Data-Level Sampling and Augmentation

Data-level methods change what the optimizer sees, rather than changing the mathematical penalty assigned to one observed example. In the taxonomy, this family includes resampling, augmentation, and WSI-specific feature mixing strategies that make minority or informative examples more visible during training (Saini and Susan, 2023).

3.3.1 Balanced Sampling

Balanced sampling changes minibatch composition by sampling training examples with probabilities inversely related to their class support. For an example i with class y_i, the sampler weight is proportional to

$$q_i \propto \frac{1}{n_{y_i}}. \tag{3}$$

After normalization across the training set, this makes the expected class exposure closer to uniform even though the underlying dataset remains long-tailed. The patch variant applies this sampler to patch examples and still optimizes CE. The MIL variant applies the same idea at slide-bag level and also optimizes CE. Balanced sampling therefore tests a pure data-level intervention: it asks whether equalizing training exposure is enough without changing the loss formula.

3.3.2 RankMix

RankMix is a WSI-native data-augmentation method for weakly supervised slide classification. Ordinary image mixup assumes two inputs with compatible spatial shapes, but WSI bags can contain different numbers of feature instances and lack instance-level labels. RankMix addresses this by mixing representative feature subsequences rather than raw slides (Chen and Lu, 2023).

The tested implementation follows the defining two-stage idea. First, a teacher attention-MIL model is trained with slide labels. For each bag $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ and target class, the teacher assigns ranking scores to instances; the highest-scoring instances are kept as a representative subsequence while preserving their original order in the bag. For two bags a and b, the selected feature matrices H_a and H_b are truncated to a shared length and mixed with $\lambda \sim \mathrm{Beta}(\alpha, \alpha)$:

$$\widetilde{H} = \lambda H_a + (1-\lambda) H_b, \qquad \widetilde{\mathbf{y}} = \lambda \mathbf{y}_a + (1-\lambda)\mathbf{y}_b. \tag{4}$$

The student MIL model is then trained with soft-label cross-entropy on the mixed bags. This mechanism is categorized as data-level augmentation because it increases the diversity of the training bags in feature space while retaining the same attention-MIL prediction architecture.

3.4 Algorithm-Level Losses

Algorithm-level methods keep the training examples fixed but alter how errors are penalized. They are cost-sensitive or difficulty-sensitive objectives: minority examples may receive larger explicit weights, or easy examples may contribute less to the gradient (Saini and Susan, 2023; Scholz et al., 2024).

3.4.1 Class-Weighted Cross-Entropy

Class-weighted CE multiplies each example's CE term by a class-dependent cost:

$$\mathcal{L}_{\mathrm{WCE}}(\mathbf{z}, y) = -w_y \log p_y, \qquad w_c \propto \frac{1}{n_c}. \tag{5}$$

The implementation normalizes the inverse-frequency weights so that the average scale of the loss remains comparable across methods. This method leaves the sample order and model architecture unchanged, but increases the loss contribution of classes with fewer training examples. It is evaluated for both patch classification and attention-MIL.

3.4.2 Focal Loss

Focal loss reduces the contribution of confidently classified examples and concentrates optimization on examples that remain difficult. With focusing parameter γ, the multiclass focal objective is

$$\mathcal{L}_{\mathrm{focal}}(\mathbf{z}, y) = -(1 - p_y)^{\gamma} \log p_y. \tag{6}$$

This study uses $\gamma = 2.0$ for the patch and WSI-bag variants. The method is relevant to imbalance because many minority-class examples can remain hard under a classifier dominated by head-class gradients, but the focal term does not directly encode the class counts. It is therefore a difficulty-sensitive loss rather than a sampler or explicit prior correction.

3.5 Hybrid Sampling–Loss Methods

Hybrid methods combine data-level and algorithm-level changes. The Scholz-style representatives are placed here because the tested variants use class-balanced oversampling together with metric-derived losses. This coupling matters: their effect cannot be attributed only to a new loss or only to a new sampler.

3.5.1 Scholz-Style CE + Soft F1

Scholz et al. motivate losses derived from metrics that are already used to evaluate imbalanced medical classification, including F1 and Matthews correlation coefficient (MCC) (Scholz et al., 2024). The intuition is that CE optimizes the log-probability of the correct class one example at a time, whereas macro F1 evaluates a different object: the balance between precision and recall at class level. In a long-tailed dataset, this distinction matters because a model can reduce CE substantially by fitting frequent classes while still leaving tail-class recall or precision weak.

The obstacle is that ordinary F1 is computed from hard counts after prediction, so it is not directly usable as a gradient-based training loss. The Scholz-style soft-F1 objective replaces those hard counts with probability-weighted counts inside each minibatch. For one class, a confident correct prediction contributes almost one soft true positive, a confident prediction for the wrong class contributes to soft false positives, and low probability on a true class contributes to soft false negatives. Formally, for one-hot labels y_i and predicted probabilities $\hat{y}_i$:

$$\mathrm{TP} = \sum_i y_i \hat{y}_i, \quad \mathrm{FP} = \sum_i (1 - y_i)\hat{y}_i, \quad \mathrm{FN} = \sum_i y_i (1 - \hat{y}_i). \tag{7}$$

These terms keep the familiar meaning of a confusion matrix but make every component differentiable with respect to the model probabilities. They then define soft precision, soft recall, and soft F1:

$$\mathrm{F1}_{\mathrm{soft}} = \frac{2\,\mathrm{precision}_{\mathrm{soft}}\,\mathrm{recall}_{\mathrm{soft}}}{\mathrm{precision}_{\mathrm{soft}} + \mathrm{recall}_{\mathrm{soft}}}. \tag{8}$$

The multiclass implementation applies this construction separately to each class and averages the resulting classwise terms. This macro-style averaging is the imbalance-aware part: each class contributes one term regardless of how many samples it has in the full dataset. The tested objective does not replace CE completely; it adds the metric-derived term to CE so that optimization still rewards calibrated class probabilities while also pushing the minibatch toward higher macro soft F1:

$$\mathcal{L}_{\mathrm{CE+F1}} = \mathcal{L}_{\mathrm{CE}} + \left(1 - \frac{1}{C}\sum_{c=1}^{C} \mathrm{F1}_{\mathrm{soft},c}\right). \tag{9}$$

The benchmark trains this loss with balanced sampling on the patch manifest. That design is important for interpretation: the loss tries to optimize a macro-style class metric, while the sampler ensures that tail classes occur often enough in minibatches for their soft-F1 terms to provide a regular training signal.

3.5.2 Scholz-Style CE + Soft MCC

The soft-MCC variant uses the same principle but starts from a different evaluation metric. MCC is often used in imbalanced classification because it summarizes all cells of the confusion matrix rather than emphasizing only the positive class or only accuracy. In multiclass form, it can be read as a normalized agreement between the predicted class distribution and the true class distribution. A high value requires more than frequent-class accuracy: the predicted mass must line up with the correct classes without collapsing onto the head classes.

To make MCC trainable, the hard predicted labels are again replaced by probabilities. Let $Y \in \{0,1\}^{N\times C}$ be the one-hot label matrix for a minibatch and $\widehat{Y} \in [0,1]^{N\times C}$ the predicted probability matrix. The numerator measures probability-weighted agreement after subtracting agreement expected from the marginal class totals. The denominator normalizes by the spread of the predicted and true class totals, so the quantity remains comparable across minibatches:

$$\mathrm{MCC}_{\mathrm{soft}} = \frac{N\sum_{i,c} Y_{ic}\widehat{Y}_{ic} - \sum_c \left(\sum_i Y_{ic}\right)\left(\sum_i \widehat{Y}_{ic}\right)}{\sqrt{N^2 - \sum_c \left(\sum_i \widehat{Y}_{ic}\right)^2}\sqrt{N^2 - \sum_c \left(\sum_i Y_{ic}\right)^2}}. \tag{10}$$

As with soft F1, the training objective turns a score to be maximized into a loss to be minimized and adds it to CE:

$$\mathcal{L}_{\mathrm{CE+MCC}} = \mathcal{L}_{\mathrm{CE}} + \left(1 - \mathrm{MCC}_{\mathrm{soft}}\right). \tag{11}$$

This variant is also evaluated only in the patch benchmark and is trained with class-balanced oversampling. The soft-F1 and soft-MCC variants therefore ask a focused question: once tail classes are regularly presented to the optimizer, does adding a differentiable version of a class-balanced evaluation metric improve the controlled patch classifier beyond CE and simple balanced sampling?

3.6 Synthetic Data Generation

Synthetic generation is data-level in spirit but is separated because it introduces new generated samples rather than resampling observed ones. In histopathology, this family is attractive because deleting majority-class tissue wastes data, while naive image-space interpolation often produces unrealistic pathology images (Saini and Susan, 2023; Ruiz-Casado et al., 2024). The practical hope is simple: if real minority-class patches are scarce, a generator may learn enough of their stain, texture, and tissue morphology distribution to provide additional plausible training examples.

3.6.1 ProGAN Augmentation

The ProGAN experiment adapts the GAN-based histopathology augmentation idea of Ruiz-Casado et al. to multiclass TCGA-UT patches (Ruiz-Casado et al., 2024). A generative adversarial network (GAN) has two coupled models. The generator G maps random latent vectors to synthetic images, while the discriminator D tries to distinguish real training patches from generated patches. Training alternates between making D better at this distinction and making G produce images that D cannot reliably reject. The standard objective expresses this competition as

$$\min_G \max_D \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{u}\sim p(\mathbf{u})}[\log(1 - D(G(\mathbf{u})))]. \tag{12}$$

This equation is not a classification loss for the final TCGA-UT classifier. It is the pretraining objective for producing additional patch images. After generation, the classifier simply sees a larger patch training set.

ProGAN modifies ordinary GAN training by growing the image resolution gradually. Instead of asking the generator to synthesize 256×256 histology patches from the start, training begins at a coarse resolution where global color and texture structure are easier to model. Higher-resolution layers are then introduced step by step until the target patch size is reached. This staged procedure is intended to stabilize synthesis and reduce the risk that the generator learns only a narrow set of repeated images. In this benchmark, class-specific generators are trained for minority patch classes, and generated patches are added until each minority class reaches the training-set head-count target. Synthetic images are used only as extra patch-classification training examples; they are not converted into synthetic WSI bags.

3.7 Representation and Architecture-Level Methods

Representation-level methods try to make rare classes more separable in feature space rather than only changing sampling or per-example loss weights. This family is especially relevant for computational pathology, where WSI labels supervise bags of many unlabeled instances and class imbalance can occur both across slides and within a positive slide (Juyal et al., 2024).

3.7.1 SC-MIL

SC-MIL extends supervised contrastive learning to MIL by applying the contrastive objective to bag representations instead of assigning noisy slide labels to individual patches (Juyal et al., 2024). The motivation is that a slide label usually does not describe every patch in a bag. A cancer-type label can be correct for the slide while many instances contain background, normal tissue, artifacts, or weakly informative morphology. Applying contrastive learning directly to instances would therefore treat many patches as positives or negatives using labels they do not truly carry. SC-MIL avoids that mismatch by contrasting the aggregated slide-level representation.

The attention-MIL backbone first maps bag B_i to an embedding $\mathbf{b}_i = \mathbf{b}(B_i)$. A projection head g then produces normalized contrastive features $\mathbf{r}_i = g(\mathbf{b}_i)$. In the contrastive view, each bag in a minibatch is an anchor. Bags with the same cancer-type label are positives because their slide-level representations should become closer, while bags from other classes act as negatives because their representations should remain separated. For anchor i, bags of the same class form the positive set $P(i)$, and the supervised contrastive term is

$$\mathcal{L}_{\text{SCL}} = -\frac{1}{\sum_i |P(i)|} \sum_i \sum_{p \in P(i)} \log \frac{\exp(\mathbf{r}_i^\top \mathbf{r}_p/\tau)}{\sum_{a \neq i} \exp(\mathbf{r}_i^\top \mathbf{r}_a/\tau)}. \tag{13}$$

The numerator rewards similarity to same-class bags. The denominator compares that similarity against all other bags in the minibatch, so the loss favors representations where class clusters are compact and mutually separated. The temperature τ controls how sharply these similarities are weighted; this benchmark uses $\tau = 1.0$.

SC-MIL does not rely on contrastive learning alone. The contrastive term shapes the representation space, while CE trains the final class predictor. The tested variant combines them through a linear curriculum:

$$\mathcal{L}_t = \beta_t \mathcal{L}_{\text{SCL}} + (1 - \beta_t)\mathcal{L}_{\text{CE}}, \qquad \beta_t = 1 - \text{progress}_t. \tag{14}$$

Early training therefore emphasizes class-structured bag embeddings, when the classifier head is still unstable. As training progresses, β_t decreases and more weight moves to CE, so the learned representation is tied back to the actual slide-level classification objective. Activation diagnostics count positive pairs during training to confirm that the contrastive component is active.

4 Evaluation

Both benchmarks use seeds 0, 1, and 2. Macro F1 is the primary ranking metric within each benchmark because it weights classes equally and penalizes failure on rare classes. Balanced accuracy, accuracy, classwise precision/recall/F1, head/body/tail summaries, negative log likelihood, multiclass Brier score, and expected calibration error are also computed where probabilities are available. Patch and WSI-bag methods are never ranked against each other, because they answer related but different questions with different input units and supervision.

All reported tables use the held-out test split and show mean $\pm$ standard deviation over seeds. Separate validation summaries are retained as artifacts for model-selection checks. The completed sweep contains no missing planned results for either benchmark.

5 Results

5.1 Patch-Level Results

Table 3: Patch-level test results over three seeds. Methods share the same patch manifest, resolution, classifier family, and evaluation code.

Method	Accuracy	Balanced accuracy	Macro F1
CE	0.412 ± 0.011	0.310 ± 0.003	0.314 ± 0.004
Focal	0.402 ± 0.010	0.307 ± 0.011	0.310 ± 0.011
ProGAN augmentation	0.407 ± 0.012	0.306 ± 0.021	0.299 ± 0.021
Balanced sampler	0.354 ± 0.011	0.333 ± 0.013	0.293 ± 0.006
Weighted CE	0.333 ± 0.033	0.330 ± 0.016	0.285 ± 0.018

Plain CE is the strongest patch method by the primary metric, with macro F1 0.314 ± 0.004 and accuracy 0.412 ± 0.011 (Table 3). Focal loss is close, with macro F1 0.310 ± 0.011. Balanced sampling and weighted CE shift the operating point toward rare-class sensitivity: they produce the highest balanced accuracies, 0.333 ± 0.013 and 0.330 ± 0.016, but lower macro F1 and accuracy. This tradeoff indicates that the class-balancing interventions improved average recall but introduced enough precision or head-class loss to reduce the harmonic classwise summary.

Among the representative advanced patch methods, ProGAN augmentation reaches macro F1 0.299 ± 0.021, below CE and focal loss (Table 3). The Scholz combined losses are reported in the same table; their macro-F1 and balanced-accuracy rankings should be read against both plain CE and balanced-sampling CE, because the Scholz configuration deliberately couples metric-derived objectives with oversampling. The result should not be interpreted as a general rejection of synthetic generation or metric-derived losses. It is narrower: under this fixed patch set, classifier family, and three-seed configuration, not every imbalance mechanism that helps in other medical imaging settings transfers to TCGA-UT patches.

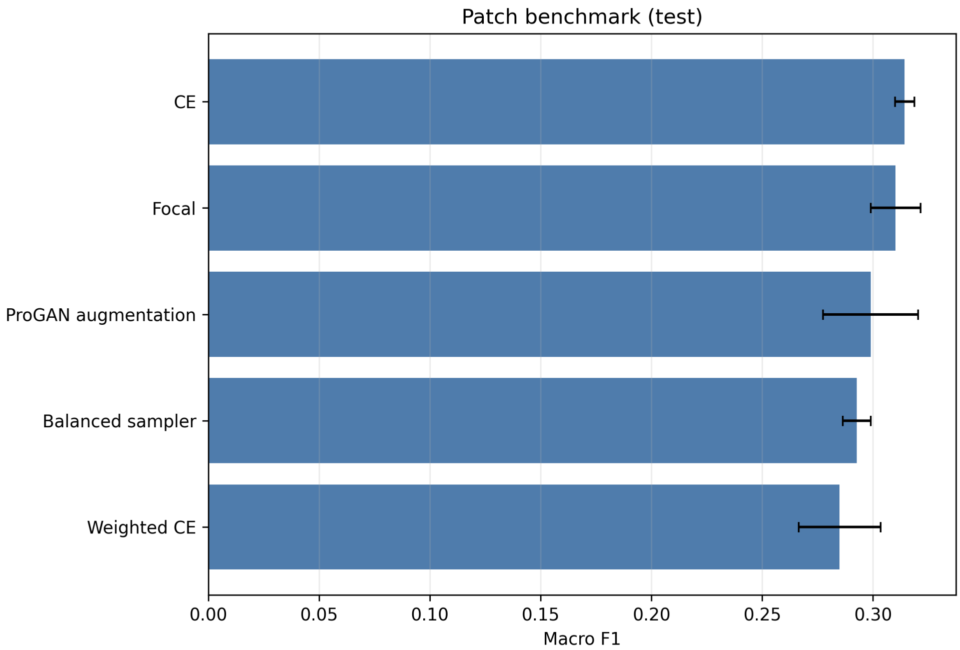


Figure 4: Patch-level macro F1 on the held-out test split. The figure visualizes the same ranking as Table 3; absolute values should only be interpreted within the patch benchmark.

5.2 WSI-Bag Results

Table 4: WSI-bag test results over three seeds. Methods share the same full feature bags, attention-MIL backbone family, and evaluation code.

Method	Accuracy	Balanced accuracy	Macro F1
RankMix	0.894 ± 0.010	0.863 ± 0.008	0.848 ± 0.017
MIL CE	0.890 ± 0.006	0.836 ± 0.020	0.840 ± 0.020
Balanced-sampling MIL	0.885 ± 0.007	0.837 ± 0.015	0.836 ± 0.013
SC-MIL	0.881 ± 0.007	0.837 ± 0.017	0.833 ± 0.016
Weighted MIL CE	0.880 ± 0.005	0.834 ± 0.018	0.829 ± 0.018
Focal MIL	0.879 ± 0.004	0.824 ± 0.011	0.827 ± 0.009

RankMix is the strongest WSI-bag method by all three aggregate metrics (Table 4). Relative to plain MIL CE, it improves balanced accuracy from 0.836 ± 0.020 to 0.863 ± 0.008 and macro F1 from 0.840 ± 0.020 to 0.848 ± 0.017. The macro-F1 gain is modest, but the balanced-accuracy gain

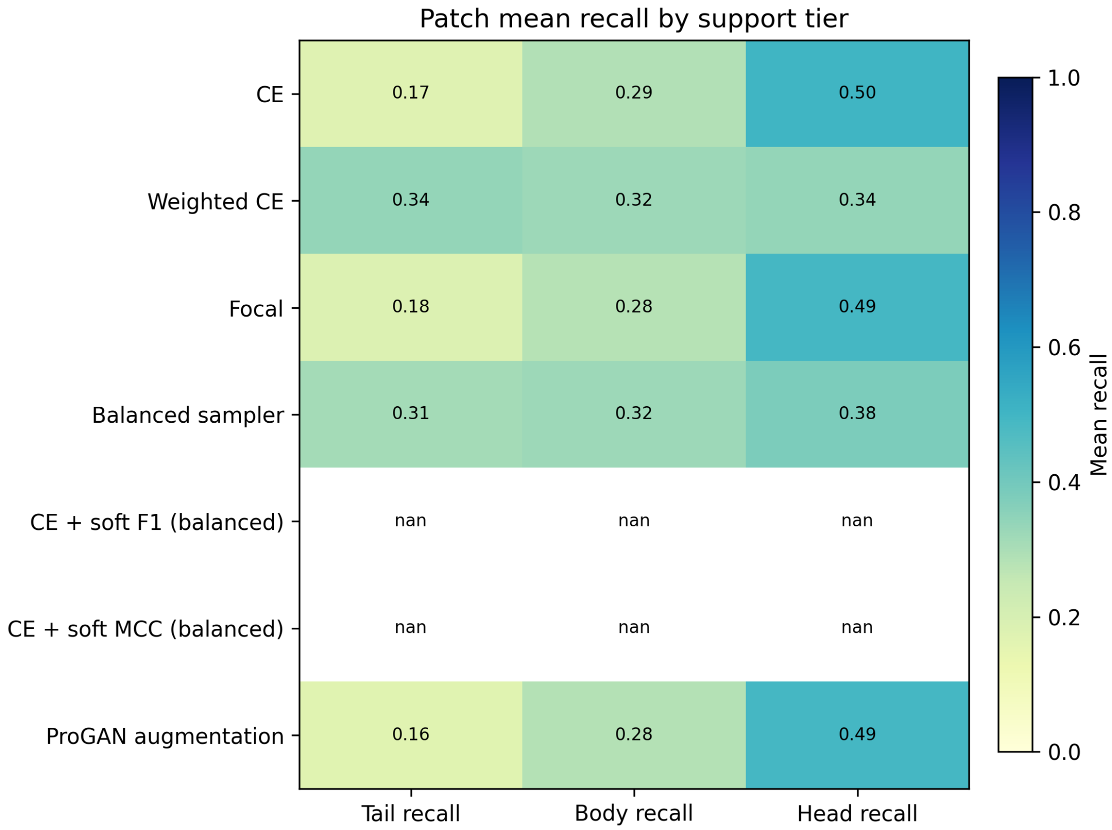


Figure 5: Patch-level classwise recall on the held-out test split. The plot exposes the class-level variation hidden by aggregate accuracy and motivates reporting balanced accuracy and macro F1.

is larger and consistent with RankMix directly augmenting representative instance subsequences in the MIL regime.

The remaining WSI methods cluster closely. Balanced-sampling MIL and SC-MIL produce similar balanced accuracy (0.837), but both trail RankMix in macro F1. SC-MIL performs well on validation but does not exceed plain MIL CE on the test macro-F1 average. Weighted MIL CE and focal MIL do not improve over plain MIL CE in this configuration, with focal MIL giving the lowest WSI macro F1. Activation diagnostics confirm that the advanced WSI objectives were active: RankMix generated about 1.10 million mixed examples per seed, and SC-MIL constructed about 1.06 million positive pairs per seed.

6 Discussion

The two benchmarks support different conclusions. In the patch benchmark, simple CE and focal loss are hard to beat by macro F1, while class-balanced sampling and weighting mainly trade macro F1 for balanced accuracy; metric-derived Scholz losses are evaluated in that same tradeoff space because they are trained with balanced sampling. In the WSI-bag benchmark, RankMix improves the strongest baseline and is the clearest case where a more elaborate imbalance-specific mechanism helps. The same study therefore mixes negative and positive transfer results across patch-level synthetic or metric-derived methods and WSI-native feature mixing.

This pattern reinforces the need to evaluate imbalance methods in their native input regimes. A patch generator should not be judged by first converting synthetic images into one-instance MIL bags, and a MIL feature-mixing method should not be compared directly against a patch

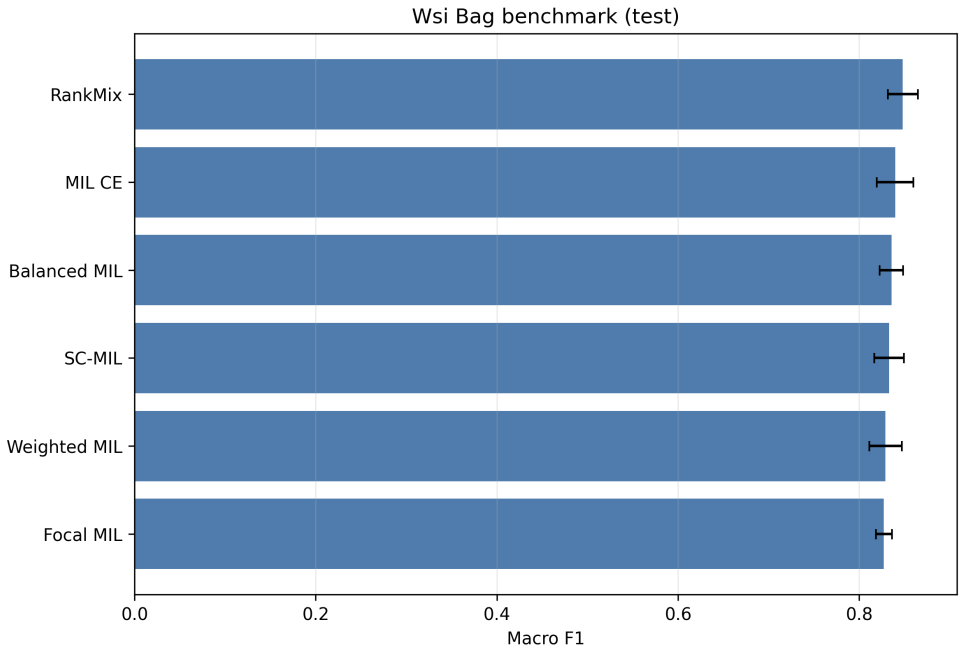


Figure 6: WSI-bag macro F1 on the held-out test split. RankMix is the top method within the WSI-bag benchmark, but the absolute scale is not comparable to the patch benchmark because the task input differs.

classifier. The controlled design narrows the claims but makes them easier to defend: within each benchmark, the split, backbone family, input budget, and evaluation code are shared.

Several limitations remain. First, the study is a controlled benchmark, not a full reproduction of every original paper. The ProGAN experiment keeps the progressive-generator mechanism from Ruiz-Casado-style histology augmentation but adapts it from binary breast histology to multiclass TCGA-UT and does not reproduce the full GAN-family comparison. Second, TCGA-UT patch labels come from pathologist-selected tumor regions; they are useful patch-level class labels but not dense annotations. Third, the WSI benchmark uses frozen Virchow2 feature bags and an attention-MIL family, so conclusions apply to this representation setting. Finally, advanced methods may benefit from additional hyperparameter tuning beyond the fixed three-seed benchmark used here.

7 Conclusion

TCGA-UT supports a cleaner imbalance study when patch-level and WSI-bag supervision are evaluated separately. On controlled patch classification, the strongest macro-F1 baseline is plain CE, and balancing interventions mainly change the recall–precision tradeoff. On WSI bags, RankMix provides the strongest overall result and the clearest improvement over a plain MIL baseline. The practical implication is that imbalance mitigation should be selected and evaluated at the same level where the model actually consumes data; otherwise, apparent method differences can be artifacts of incompatible task regimes.

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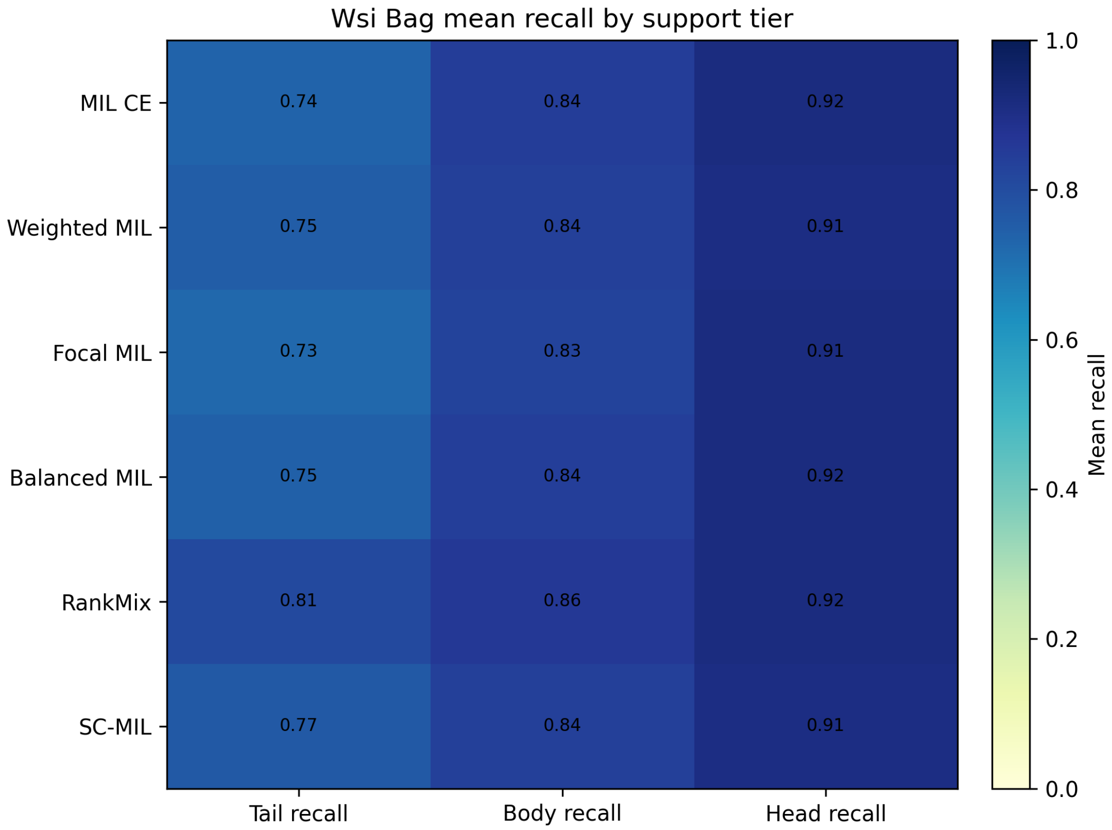

Figure 7: WSI-bag classwise recall on the held-out test split. The spread across classes shows why aggregate accuracy alone is insufficient for the long-tailed setting.

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